# Homework Exercise Sheet 4
# Introduction to Cryptography & IT-Security
# PS 511.088
# Summer Semester 2026

YOUR-GROUP-NAME-FROM-BLACKBOARD-ONLY-HERE-NO-NAMES

May 30, 2026

This is the forth homework exercise sheet for the proseminar *Introduction to Cryptography and IT-Security* (PS 511.088) covering the block *Classical Cryptography and Secure Protocols* to be completed before the proseminar on June 9, 2026 with a maximum of 20 points to be achieved. Submission deadline in Blackboard is June 2, 2026 at 12:59 PM. Late submissions will **not** be graded.

Use CrypTool software and/or SAGE to assist you in solving the exercises and the provided LATEX code in the appendix to format your report, using the source LATEX of the homework exercise sheet. Submit both resulting `tex/pdf` files in Blackboard. No other format is allowed.

# PS-B4-E1: Diffie-Hellman (DH)

The Diffie-Hellman key exchange (DHKE) is among the first published works on public key cryptography and has been integrated into many protocols. Its underlying mathematics are important to understand for their use in this cryptographic application of group theory.

**PS-B4-E1: DH-Q1: Practical computing with DHKE**

You are intercepting a connection between Alice and Bob and received the public Diffie-Hellman prime and generator

- $p = 37$
- $g = 6$

Next, you observe that Alice sends Bob the number $a = 36$ and Bob sends $b = 31$ to Alice.

1. Find the key on which they agreed.
2. What makes the recovery of this key so easy?
3. From Alice's and Bob's viewpoint, what would be a better choice of $g$?

5 pts

**Answer: YOUR-SOLUTION-HERE**

**PS-B4-E1: DH-Q2: Practical computing with DHKE**

In the Diffie-Hellman key exchange, $g$ is chosen so that $2 \leq g \leq p - 2$. Give arguments why the values:

- $g = p - 1$,
- $g = p + 1$,

are good or bad choices, respectively?

1 pts

**Answer: YOUR-SOLUTION-HERE**

**PS-B4-E1: DH-Q3: Practical computing with DHKE**

Digital signatures can be used to prevent a man-in-the-middle attack on Diffie-Hellman. Suppose that Alice and Bob already share a symmetric key $K$ before they begin the Diffie-Hellman procedure. Draw a diagram to illustrate a simple method Alice and Bob can use to prevent a man-in-the-middle attack.

2 pts

**Answer: YOUR-SOLUTION-HERE**

# PS-B4-E2: Digital Signature Standard

The Digital Signature Standard (DSS) was published by NIST and has been officially replaced. It represent *early efforts* to advocate and push for widespread use of standardized cryptography.

**PS-B4-E2: DSS-Q1: Attack on optimized operational use of DSS**

Consider the DSS with the following parameters:

- prime $q$;
- prime $p = aq + 1$ for some $a \in \mathbb{Z}$;
- element of order $q$ in $\mathbb{Z}_p^*$;
- a cryptographic hash function H;
- a secret key $x \in \mathbb{Z}_q$.

Consider the following mode of operations for applying DSS (in violation of the standard):

- A User creates public/private key pair.
- This user selects $k$ as required and further computes a pair $(k, r)$ such that $r \equiv \left(g^k \mod p\right) \mod q$
- This user **always** uses this pair (i.e., these two values $k$ and $r$) to compute any signatures.

Assume that you have obtained - at least two - signatures that have been produced as described above. Show that in this case it is possible to perform an attack leading to universal forgery capabilities for the attacker by recovering the secret key $x$.

3 pts

**Answer: YOUR-SOLUTION-HERE**

**PS-B4-E2: DSS-Q2: Attack on optimized operational use of DSS**

In the context of the previous question, argue why the adversary has obtained universal forgery capabilities for signatures. List the concrete steps using an example.

0.5 pts

**Answer: YOUR-SOLUTION-HERE**

## PS-B4-E2: DSS-Q3: Attacking RSA signature

You have intercepted the following message and RSA signature:

- 150 dollars
- 56840167481593713576269217621738921774093

The signature was generated by using the str2num command of Sage to turn the message into an integer $m$, using the following public key $(n, e)$:

- $n = 2^{137} - 1$
- $e = 17$

Show how to use the **factor** command of sage to break this system and generate a forged signature for the following message:

- 250 dollars

Demonstrate the forged signature using CrypTool 1: Menu Digital Signatures/PKI ➞Signature Demonstration (Signature Generation) ...

3 pts

**Answer:** **YOUR-SOLUTION-HERE**

## PS-B4-E2: DSS-Q4: Theoretical aspects on RSA signature attack

In the context of the previous question, argue why the adversary has obtained partial or universal forgery capabilities for signatures. List the concrete steps using an example.

0.5 pts

## PS-B4-E3: Transport Layer Security

The Transport Layer Security protocol is fundamental in modern network architectures and a major building block for modern Internet. Defining, implementing and deploying TLS is very complicated and there have been many security issues found in the past. The different versions of TLS have been specified in several RFCs that also point to other data such as the official TLS Cipher Suites list published by IANA (`https://www.iana.org/assignments/tls-parameters/tls-parameters.xhtml#tls-parameters-4`).

### PS-B4-E3: TLS-Q1: TLS Cipher Suites

Consider the IANA list of official TLS Cipher Suites.

- Do all listed ciphers provide security?
- Do some ciphers provide 'more' security than others?

0.5 pts

**Answer: YOUR-SOLUTION-HERE**

### PS-B4-E3: TLS-Q2: TLS Cipher Suites

A paranoid client wants to connect to a TLS server. For some reasons, the client prefers to avoid cryptographic standards from the US Government and would like to rely on symmetric keys of at least 128 bits for his very secret transaction.

```
TLS_NULL_WITH_NULL_NULL
TLS_RSA_WITH_NULL_MD5
TLS_RSA_WITH_NULL_SHA
TLS_RSA_EXPORT_WITH_RC4_40_MD5
TLS_RSA_WITH_RC4_128_MD5
TLS_RSA_WITH_RC4_128_SHA
TLS_RSA_EXPORT_WITH_RC2_CBC_40_MD5
TLS_RSA_WITH_IDEA_CBC_SHA
TLS_RSA_EXPORT_WITH_DES40_CBC_SHA
TLS_RSA_WITH_DES_CBC_SHA
TLS_RSA_WITH_3DES_EDE_CBC_SHA
TLS_DH_DSS_EXPORT_WITH_DES40_CBC_SHA
TLS_DH_DSS_WITH_DES_CBC_SHA
TLS_DH_DSS_WITH_3DES_EDE_CBC_SHA
TLS_DH_RSA_EXPORT_WITH_DES40_CBC_SHA
TLS_DH_RSA_WITH_DES_CBC_SHA
TLS_DH_RSA_WITH_3DES_EDE_CBC_SHA
TLS_DHE_DSS_EXPORT_WITH_DES40_CBC_SHA
```

```
TLS_DHE_DSS_WITH_DES_CBC_SHA
TLS_DHE_DSS_WITH_3DES_EDE_CBC_SHA
TLS_DHE_RSA_EXPORT_WITH_DES40_CBC_SHA
TLS_DHE_RSA_WITH_DES_CBC_SHA
TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA
TLS_DH_anon_EXPORT_WITH_RC4_40_MD5
TLS_DH_anon_WITH_RC4_128_MD5
TLS_DH_anon_EXPORT_WITH_DES40_CBC_SHA
TLS_DH_anon_WITH_DES_CBC_SHA
TLS_DH_anon_WITH_3DES_EDE_CBC_SHA
TLS_RSA_WITH_AES_128_CBC_SHA
TLS_DH_DSS_WITH_AES_128_CBC_SHA
TLS_DH_RSA_WITH_AES_128_CBC_SHA
TLS_DHE_DSS_WITH_AES_128_CBC_SHA
TLS_DHE_RSA_WITH_AES_128_CBC_SHA
TLS_DH_anon_WITH_AES_128_CBC_SHA
TLS_RSA_WITH_AES_256_CBC_SHA
TLS_DH_DSS_WITH_AES_256_CBC_SHA
TLS_DH_RSA_WITH_AES_256_CBC_SHA
TLS_DHE_DSS_WITH_AES_256_CBC_SHA
TLS_DHE_RSA_WITH_AES_256_CBC_SHA
TLS_DH_anon_WITH_AES_256_CBC_SHA
```

1. Select the only two cipher suites from the list above which satisfy the security policy of the client. Notice that one of the two requires to authenticate a public key. Identify which one.
2. Consider the cipher suite which requires public key authentication. Assume that an X.509 certificate is transmitted from the server to the client, but that the certificate authority is not known by the client. What happens? What is the typical reaction of the user?
3. Compare the security of the two cipher suites. Are they considered secure today?

3 pts

**Answer:** **YOUR-SOLUTION-HERE**

## PS-B4-E3: TLS-Q3: openSSL message signing

You have already seen how the signing process with RSA works using Cryptool 1 in a previous task. Now you are going to do this procedure using the openSSL utility in Cryptool Online. You can use the GUI or the shell commands directly.

1. Go through the procedure of generating a key and and then sign the message "Hello world!" and the file signme.txt that is provided in the exercise materials and document your process.
2. Compare the generated signatures. Are the signatures the same?
3. Now use openssl to verify the signatures and document your process.

1.5 pts

**Answer: YOUR-SOLUTION-HERE**

# Appendix: Instructions for Homework Sheet Preparation in LaTeX

## 0.1 How to Include an Image

A picture can be included in your document using the LaTeXcode used to obtain Figure 1.

## 0.2 How to Include a Table

A table can be included using the commands used to obtain Table 1.

| Software | Assessment | Comments |
|---|---|---|
| Suite A | works well | internal use only |
| Suite B | freely usable | observe guidance document |
| Suite C | restricted | Requires compiler |

Table 1: Caption

## 0.3 How to List Data like Hexdumps

Hexadecimal encoding provides a way to encode data in a machine and human-readable way. The code that has been used to generate Listing 1 can be used to include hex-data, including the one provided by CrypTool1.

Listing 1: Example for how to include hex-data.

```
50 63 D8 D0 D8 E4 FC F7 09 FB 1F 40 C1 99 E7 62 56 8A 20 77 B9 D6 F7 21 B3 61 F0 8D
96 AA 45 E3 A0 A5 11 97 58 17 2B 53 60 6D A3 2C A0 4C 2E 1F 5F 31 94 6A CC 09 28 8E
E0 0E 7C 23 2E 2F E3 E6 6A 02 AA D2 A7 41 80 6D AF 5D 72 63 0E AE A8 1F 3A 38 E3 91
74 56 B6 D7 59 C8 14 4B 0F C2 D8 9F 46 98 26 92 AF 5D 60 9E 5D AC 30 A1 DD 7A D1 95
49 AD 54 3C 93 B0 DF BC B2 57 83 AF F7 43 D8 98 0B 06 A4 6B 42 4C 8D 44 F4 7C BB B0
A6 B3 A2 7E 9C 11 D7 A6 98 23 EC 23 8E C8 AC FD 15 B0 2F 2B EC 49 25 B5 70 9C 0B D5
8A 24 5F 8C 7D 1A 32 E9 4C EF E7 B1 06 C7 D4 A2 C8 DA 01 15 99 E2 92 6B 0E DF 5B 36
15 2B 30 30 E0 5A A6 E0 80 2F 07 4F B4 4F 66 F8 17 A0 A3 72
```

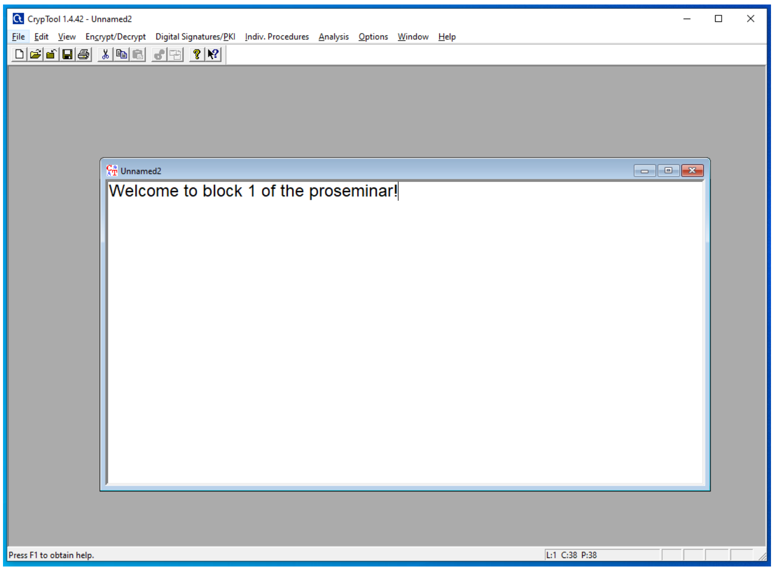


Figure 1: The code of this figure can be used when preparing your homework.